

Lost in Translation: A Detrimental Impact of Bilingualism in The Testing Effect

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Introduction:

Students in Wales sometimes learn in one language however, instruction occurs in either Welsh or their native language. As they move to university, English becomes the medium of instruction and learning.

This change has shown to be detrimental to measured performance however does this still occur during testing of the material?

The Testing Effect (TE) (Abott, 1909) is a robust paradigm indicating testing is more beneficial than restudying. TE requires increased cognitive effort and attention similarly to bilingualism. Bilingualism promotes similar advantages to TE, and could follow the same mechanisms.

2016 PISA found children who learnt in Welsh and tested in English outperformed those taught and tested in one language, however, we are not aware of a study that has yet to directly examine the impact of studying in one language & being tested in the second language.

Objective:

Is the Testing Effect evident across different languages?

Hypotheses

- No performance difference or a decrease in performance across conditions
 - Transfer Appropriate Processing*
- Increase in performance across conditions
 - Linked to the Retrieval Effort Theory due to increased cognitive effort of learners, thus improving learning*

Method:

40 participants:

- Initial study phase:** 20-word pairs
Parity task (30 seconds)



- Learning Phase:** Restudy OR Test
Parity task (30 seconds)



OR



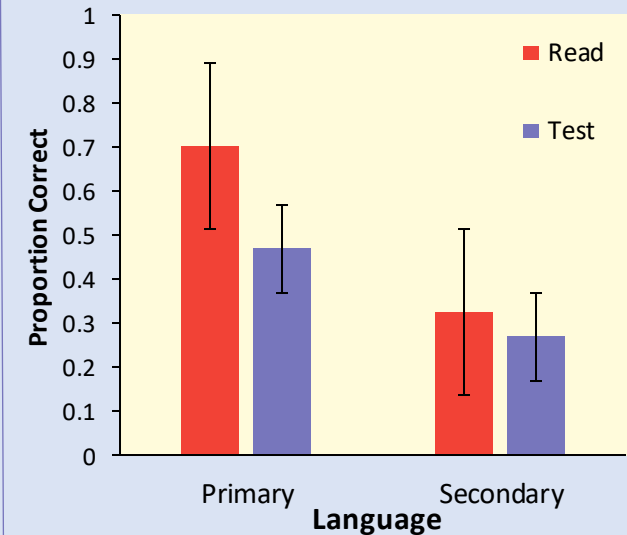
- Final test:**
 - L1 (Primary Language)
 - OR
 - L2 (Secondary Language)



OR



Results:



Discussion:

There is an impairment in the proportion of correct answers when switching languages which show a greater effect of Primary language between Read and Test conditions compared to secondary.

The Testing Effect had a detrimental effect in our study. However, we did not accurately replicate TE as the delay between study and test was not long enough.

Implications – what learning strategy can mitigate the risk of decreased performance when switching languages during learning and testing?